Available online at www.sciencedirect.com

ScienceDirect

International Journal of Project Management 33 (2015) 481 – 482

International Journal of Project Management

www.elsevier.com/locate/ijproman

Editorial

Management of Infrastructure Projects for Urbanization in China

1. Introduction

Infrastructure projects are among the main triggers for urban development. China is experiencing an important historical period of urbanization in which many infrastructure projects are under construction or planned for construction. This special issue attempts to address the main topics on how to manage infrastructure projects, particularly from the perspective of how to align the management of infrastructure projects with urbanization in China. Although the development of infrastructures may create great opportunities for urbanization, the management of infrastructure projects usually faces the challenges of the tradeoff between multi-objectives tradeoff, the involvement of multi-stakeholders, and uncertainties in the dynamic environment. Therefore, it is necessary to explore adaptable theories, methodologies, tools and techniques in project management domain to cope with uncertainty, ambiguity and fuzzy issues and find more robust solutions for managing infrastructure projects.

2. Comments on papers

In this special issue, 11 papers are compiled from researcher in China. All of them address research issues relevant to management for infrastructure projects in the context of China.

Qian Shi, Yang Liu, Jian Zuo, Ningxia Pan and Guofeng Ma explore how to manage social risks of infrastructure projects, particularly during the process of urbanization in China. A case study of hydraulic project was employed to demonstrate the identification of social risks, involving the understanding of related stakeholders, the assessment of social risks based on observations, expert meetings, interviews and discussion forums. Response plans were developed to prevent, mitigate and cope with the potential consequences of social risks that may occur before or during the implementation of infrastructure project.

Shuibo Zhang, Ying Gao, Zhuo Feng and Weizhuo Sun examine a question of how to cultivate a facilitating institutional environment for developing PPP projects success, focusing on the institutional analysis of the Chinese context. Based on the establishment of a theoretical framework of PPP governance, which presents the logics and interactions of the differentiating levels in the PPP system, the research introduces the institutional framework with Chinese context with regard to its cultural, legal, and administrative characteristics. The application of the framework can enhance China's institutional arrangements for healthy PPP promotion.

Yongkui Li, Yujie Lu, Young Hoon Kwak, and Shuang Dong present the development of a city-level multi-project management information system to decompose the information processing complexity in the context of agent construction model (ACM). The complex adaptive system and two specific development techniques, namely, adaptive project framework and modularized functional design method, are applied for the system development. The system was validated in a typical urbanization city, Changchun, China. This research complements the existing project information system by adopting complexity design principles and it provides alternative method in managing information for large-scale urbanization projects.

Liyin Shen, Liudan Jiao, Bei He, and Lanchun Li introduce a model for evaluating the utility efficiency of metro infrastructure projects (MIP) with reference to the Chinese context. The development of metro infrastructure projects (MIP) has been playing an important role in the economic growth in China through improving urban transportation condition. The concerns on the utility efficiency and the impacts of MIP on social and environmental aspects have been increasingly raised. This research provides an alternative method to guide the development of MIP towards achieving better utility efficiency and collective benefits between economic, social and environmental dimensions.

S.X. Zeng, H.Y. Ma, H. Lin, R.C. Zeng, and Vivian W.Y. Tam examine the social responsibility in implementing major infrastructure project. The key issues of major infrastructure projects' social responsibility (MIP-SR) are investigated by developing a comprehensive conceptual framework. The framework presents in three dimensions: project life-cycle dynamics; stakeholder's heterogeneity, and social responsibility interactivity.

Qinghua He, Lan Luo, Yi Hu, and Albert P.C. Chan develop a model for measuring the complexity of mega construction projects by using fuzzy analytic network process technique. The study was based on the Shanghai Expo construction project in China. In this measurement model, factors contributing

http://dx.doi.org/10.1016/j.ijproman.2015.01.011
0263-7863//00/© 2015 Elsevier Ltd. APM and IPMA. All rights reserved.

to the complexity are grouped under six categories, namely, technological, organizational, goal, environmental, cultural and information complexities. The research work provides valuable references for managing megaprojects.

Junying Liu, Fanye Meng, and Richard Fellows examine how culture influences contractors' risk management through four practical cases selected in China, Poland and Singapore. A conceptual framework is developed for understanding the relation between culture and risk management. The findings show that project risks are perceived and managed differently when people work in the environments with different cultures. It further supports that contractors' knowledge of the host country's national culture influences their risk management behaviors.

Xiaolong Xue, Ruixue Zhang, Xiaoling Zhang, Rebecca Jing Yang, and Hong Li analyze the key factors for measuring environmental and social influences of subway construction and their interrelationships by using structural equation modeling (SEM) method, with reference to the Chinese context. They argue that, whilst subway projects can relieve transportation pressure in cities and make citizen's life easier, this type of projects also causes various environmental and social problems during the construction process. The research provides valuable method for assessing key environmental and social impacts of urban subway constructions, thus proper construction programs can be adopted to minimize negative impacts of subway construction in the Chinese urbanization development process.

Pengcheng Xiang, Xiaosen Huo, and Liyin Shen argue that asymmetric information leads to opportunistic behavior, which is the primary cause of loss and risk in the construction market. They analyze the probability and impacts of asymmetric information phenomena in the process of managing construction projects with reference with the Chinese context. The research focuses on information asymmetry among owners, contractors and project management supervisors. The research helps to distinguish the primary and secondary order of asymmetric information phenomena and improve the efficiency of project management.

Liang Liu, Chuanfeng Han, and Weisheng Xu appreciate that construction organizations are increasingly cooperating to share risks, pool resources and explore opportunities to jointly participate in managing large-scale projects. They build contractors' collaboration networks by mapping the electronic database of National Quality Award Projects of China (NQAPC) for an 8-year period (2003–2010). They investigate the structural evolution of the collaborations between contractors in the construction industry of China. The research findings help both administrative departments understand the social mechanisms driving the operation of construction industry, and contractors choose competent partners by identifying network properties.

Yunbo Lu, Lan Luo, Hongli Wang, Yun Le, and Qian Shi appreciate that measuring project complexity has become an integral part of project management as large-scale projects have grown in size, quantity, and complexity. They introduce an alternative measurement model for measuring project complexity from task and organization (TO) perspective. By using the case of Shanghai World Expo construction project, the research demonstrates that the proposed measurement method can be used to reflect effectively the project complexity.

3. Conclusion

This special issue starts a new journey for the research on infrastructure projects management by refining the issues that need to be urgently addressed. It is the guest editors' delight to assemble these eleven papers and we believe that this work provides insights into the diversity of research interests in the Chinese project management community. We would like to thank the IJPM team for the publication of this special edition.

Liyin Shen
Professor, Guest Editor
Faculty of Construction Management and Real Estate, Chongqing University, Chongqing, China
Corresponding author.

Qian Shi
School of Economics and Management, Tongji University, Shanghai, China